PPG Theoretical Knowledge Test

SAHPA
SOUTH AFRICAN HANG GLIDING
AND PARAGLIDING ASSOCIATION

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_____/150 = ____100____% PASS / FAIL (requires 85% min to pass)

You are encouraged to complete this paper as soon as convenient, after or during your course, whilst all information is still fresh in your mind.

Air law questions to be passed, 100%

NOTE: This is an open book research examination. Please answer concisely. The test answers must be your own effort, and should not be copied from others!

Draw sketches where required (distinguish between describe and/or illustrate).

Where sketches are provided, write answers on them.

All Answers must be typed out, NO handwritten Answers and must be in English

Question 1: (PTE) **(10)**

Explain what you understand about the ***Propeller Torque Effect***, including its **causes** and **effects** and how it **affects PPG flight**. What aspects influence the **extent** of the effect, and **steps to be taken** to minimise the consequences.

If the prop spins anti-clockwise when viewed from behind as with most pusher props, the engine and frame is forced to twist and rotate clockwise because of blade drag plus inertia of the prop, crank and flywheel, especially during acceleration.

The effect of this torque will cause the frame to pull down on the right risers and lightens the load on the left risers, acting like a weight-shift and causing a turn to the right.

To reduce or minimise the consequences one can:

- offset hang points
- diagonal harness-to-seat-board connection
- trim offset
- counter-rotating prop

Question 2: (GP) **(10)**

Describe the **Gyroscopic** Aspects, with particular emphasis on ***Gyroscopic Precession***. What it is, what **causes** it and how it **affects PPG flight** and the resulting **dangers** thereof. Detail some **examples**, including resulting action and powered paraglider **behaviour**.

A spinning propeller acts as a gyroscope and resists a change of its spin axis. If a force does change that axis, the resulting motion appears 90° later in the direction of rotation (precession).

Example: If your prop rotates counter-clockwise (as viewed from behind), if you pitch up (lean back) the motor yaws sharply left and you can end up facing the left wingtip.

The resulting danger is if you are close to the ground in a turn with throttle, because of the unexpected yaw, you can lose height or be pointed towards the ground.

Unlike Asymmetric Blade Thrust, gyroscopic precession is momentary and happens only when pitch or yaw is changing. You can reduce power if it surprises you.

Question 3: (ABT) **(15)**

Describe the **three causes** of ***Asymmetric Blade Thrust*** and its **effect**. What **factors** influence the extent of the problem and how can it be reduced.

Asymmetric Blade Thrust occurs when the propeller disc is not perpendicular to the relative airflow (usually tilted back in flight).

Three contributing causes:

1. The descending blade (left side, viewed from behind) has a higher angle of attack than the ascending blade, so more thrust on that side → yaw.
2. The descending blade also has a higher relative airspeed, adding still more thrust on that side.
3. The pilot's body blocks airflow; as the motor yaws, cleaner air from one side increases the yaw and can cause oscillation.

It is almost constant while the disc is tilted (unlike gyroscopic precession). Worse at high power and with a laid-back hang angle. Reduce it by:

- a more upright hang
- less power
- correct hang-point setup

Question 4: (Attachments) **(5)**

Explain why powered paragliders usually have higher ***Attachment Points*** (riser attachments) than free-flight harnesses. Describe the various **methods of attachments** and the **advantages- and disadvantages**- of each.

Powered paragliders usually have higher hang points, above the motor's centre of gravity, to make the wing pivot from a higher point, so less turbulence is transferred to the pilot, launches are generally easier, and you can sit more upright. The trade-off is less weight-shift and a longer reach to the brakes.

Types of attachment (and pros/cons):

High - carabiners above the shoulders.

- *Advantages:* most stable in turbulence; often easier to launch; can be set more vertical, which reduces torque.

  Soft high = slight weight-shift; hard high = almost none.

- *Disadvantages:* poor weight-shift (more brake/tip steering); arms stretch for brakes; if you hang too far back you can still get a strong torque twist.

Mid - Modern gooseneck / S-arm put the carabiners near the thrust line.

- *Advantages:* usual compromise; gooseneck gives useful weight-shift with less power-induced pitch. Most popular on current machines.
- *Disadvantages:* not as "free-flight" as a true low; not as isolated from bumps as a high.

Low - near the pilot or below the motor's CG, usually on pivoting arms. Meant to feel like free flight.

- *Advantages:* strongest weight-shift and wing feedback; lower, more comfortable brake position.
- *Disadvantages:* more bumpy in turbulence; harder for new pilots to launch; more lean-back and torque (especially older non-S-arm); full power can pitch you forward; toggles may need adjustment for a full flare.

Question 5: (running-in) **(10)**

Discuss ***running-in*** of new engines, the **purpose** thereof, **procedures** and **precautions**. Cover aspects of **carb tuning**, ratio of **oil** mixtures, **vibration** problems, **loosening** fittings, etc

A new engine needs to be run in so the piston, rings and bore can bed in as they heat up and cool down. That helps the engine last. Always follow the owner's manual or the manufacturer's specification. A typical run in is done as three 15-minute cycles.

On the carb, start rich and lean the high-speed screw until you get peak rpm. Too lean can burn the engine; too rich gives low power and a fouled plug. Set idle with the idle screw.

Use a richer oil mix for the first 10 hours or so (often 3%, or 30 ml per litre). After that, go to the manufacturer's mix, usually about 2%.

Check the motor before and after every flight. Look at bolts, springs and engine mounts, loose items like the airbox or filter, the throttle cable, the fuel system for leaks, and the propeller bolts and tracking. Loose fittings will vibrate and work themselves loose.

Question 6: (Fuel /oil) **(5)**

Comment on ***Fuel Mixtures*** with specific reference to **oil types** (i.e. Synthetic vs Mineral oils), mixture **ratio's** and **running-in** considerations.

Use only the oil the manufacturer specifies. For almost all 2-stroke paramotors that means full synthetic, because it handles heat better than mineral oil. Mineral oil has low heat tolerance; semi-synthetic is a blend. Mix in a separate can: put the oil in first, then the petrol, close the cap and shake it well. Typical ratios in the notes are 40:1 (250 ml per 10 L) or 50:1 (200 ml per 10 L).

Run in is richer for the first 10 hours or so (often 3%, or 30 ml per litre). After that, go to the manufacturer's mix, usually about 2%.

Question 7: (Heat Seizure) **(15)**

Discuss ***heat-seizure*** of engines, **causes** thereof, assessing **extent of** damage, resulting **repairs** required, and **preventative** action.

Heat seizure is when the piston gets so hot it expands and grabs in the bore. The engine can stop dead. Sometimes the crown even melts and a hole burns through the top of the piston. Paramotors are air-cooled and run hot, so they are easy to seize if something goes wrong.

Causes
The usual cause is the engine running too lean (too much air, not enough fuel). Combustion then gets much hotter than it should. Typical reasons:

- carb set too lean, or a blocked fuel filter / fuel pump / tank vent so the engine is starved
- an air leak (loose head bolts, cracked intake)
- not enough cooling airflow (static full throttle or long idle on the ground)
- wrong oil, or too much oil (the mix gets thicker, the carb leans out, and it runs hotter)
- cheap mineral oil that burns and stops lubricating

Once it is too hot, the oil on the piston burns, the ring can stick, and the piston expands until it seizes.

Assessing the damage
Take the top end off and look at the piston, rings and bore. Burnt black oil on the piston, scuff or seizure marks on the skirt, a stuck ring, or a hole in the crown all mean it overheated. Light scuffing may clean up. Deep scores, a melted crown or a ruined electroplated coating is serious.

Repairs
You need a new piston and rings as a minimum. A cast-iron sleeve can often be honed or bored and fitted with an oversize piston. A plated (Nikasil) bore often cannot be bored — it may need a new cylinder or specialist replate. Fix whatever caused it before you run it again.

Prevention
Use the manufacturer's full synthetic oil and the correct mix — do not just add extra oil. Keep the fuel system clean. Warm the engine before full throttle. Do not sit on full power or long idle with no airflow. A CHT gauge gives you warning before it seizes. If it does stop in the air, treat it as an engine-out and land.

Question 8: (SC Rule) **(3)**

Briefly describe the concept of the ***Semi-Circular Separation Rule***, as pertaining to air-traffic separation. (no need to reproduce the actual table here)

The semi-circular rule keeps aircraft apart vertically. It only applies above 1500 ft AGL in uncontrolled airspace.

You set QNE and choose a flight level from your magnetic track:

- 000° to 179° (easterly) - odd flight levels + 500 ft (for example FL075)
- 180° to 359° (westerly) - even flight levels + 500 ft (for example FL065)

VFR is only allowed up to FL195.

Question 9: (Line Rule) **(2)**

Explain the principles and application of the ***Following Line Rule***.

When flying at or below 1500 ft AGL and within 1 NM of a line feature (road, power line, railway, coastline), fly to the RIGHT of that line. The rule applies in Class F and G (uncontrolled) airspace.

Question 10: (Abbr.) **(16)**

What do the following ***abbreviations*** stand for, and give a brief description of each:

1. VMC

Visual Meteorological Conditions - weather in which you fly by visual reference to the ground, meeting the VMC minima.

2. VFR

Visual Flight Rules - the rules for flight in VMC (see-and-avoid, cloud/visibility limits, below FL195).

3. QNE

Standard pressure setting 1013.25 hPa. With QNE set, the altimeter reads Pressure Altitude / Flight Level.

4. AGL

Above Ground Level - height above the terrain beneath you.

5. FL

Flight Level - altitude in hundreds of feet on QNE (e.g. FL075 = 7500 ft on 1013 hPa).

6. UTC

Coordinated Universal Time (Zulu) - the worldwide aviation time standard.

7. CTR

Control Zone - controlled airspace (usually Class C in SA) from the surface around an aerodrome.

8. FAR

FA Restricted Area (South Africa) - named FA + R. Civilian aircraft may enter only with permission from the controlling authority (usually military).

Question 11: (Radio broadcasts) **(2)**

How often do you need to make general radio announcements while in flight?

Broadcast intentions about every 5–10 minutes (TIBA) and keep a listening watch on the correct frequency for the area.

Question 12: (Radio broadcasts-a) **(6)**

Write an example of a general radio broadcast announcement for your area.

“Traffic in the Special Rules Area West, this is PPG-DYB, powered paraglider, 3 miles south of Atteridgeville Quarry over grasslands, 5000 feet and below for the next 30 minutes. Traffic in the area, slow-flying aircraft, powered paraglider, PPG-DYB.”

(the area, who you are, aircraft type, position, altitude, what you plan to do, then your callsign again)

Question 13: (motor inspections) **(4)**

When should you inspect your motor for potential problems (nuts, bolts, rubbers, belt, etc) ?

As often as possible - a minimum before AND after every flight (prop, cage, bolts, rubbers/mounts, belt/gearbox, throttle, fuel system, hang points).

Question 14: (kill switch) **(3)**

If you are about to land and discover that your engine kill-switch is not functional, what would you do and how would you kill the engine?

Fly and set up a normal landing. Kill the engine by

- pinching/bending the fuel line
- covering the air intake
- Use the master switch

Then land and stay away from the prop.

Question 15: (Speed Bar) **(2)**

Why is it important to have your speed-bar fitted?

The speed bar lowers the wing's angle of attack so you go faster and cut through the wind better. If the wind picks up and you are barely moving forward, you push the bar to get penetration and still reach a landing field.

Question 16: (Trim-tabs) **(2)**

What are the benefits and usages of having trimtabs on a powered wing?

Trims change the wing's speed (with the speed bar). You can also adjust trims to help with propeller torque. On a reflex / student PPG wing, closed to neutral trims keep it slow and easy. Open / reflex trims make it faster.

Question 17: (formation flying-a) **(2)**

If you are flying with a friend, what must you beware of?

Always know where the other pilot is. Stay well apart and don't sit at the same height.
A mid-air or his prop in your lines can wreck the wing. Keep clear of his prop wash.
Use the radio if you have one.

Question 18: (Formation Flying-b) **(2)**

If you are flying alongside a buddy and you accidentally bump wingtips, what is the correct course of action for both pilots?

Both pilots should turn away from each other if it is safe to do so, restore control, then continue well separated.

Question 19: (over water) **(2)**

If you are flying near the coast (or lake or dam), what is a safe distance and height to go out over the water?

Only as far and as high as your glide ratio will still let you reach land after an engine-out.

Question 20: (Log book) **(1)**

Why would you log your flying hours?

a) to record progress
b) for licensing and renewal purposes
c) proof of experience and licenses received
d) to keep track of engine hours for servicing intervals
e) all of the above.

Question 21: (safety altitude) **(1)**

If you double your flying height above the ground, by what factor is the landing area you can reach by gliding increased? (in the event of an engine-out)

a) Double

b) Triple

c) Four times

d) 50%

Question 22: (Clouds) **(4)**

How far do you need to remain away from clouds? Vertically and horizontally

We fly under Visual Flight Rules, you must see the ground, not sit under more than 3/8 cloud cover within 5 NM, and we must meet the VMC minima table.

According to the VMC minima table the distance from clouds are:

- Vertically: 500ft
- Horizontally: 600m

Question 23: (sunset flying) **(1)**

When flying around sunset, what is the latest you may remain in the air?

We can remain in the air at the latest 15 minutes after sunset.

Question 24: (Tandem flying) **(3)**

What do you need before you may take a passenger for a tandem flight on your paramotor? (in terms of licensing and equipment)

One must hold a tandem rating and only tandem from a Director-approved site.

Licence

- Valid SACAA / PPG licence and SAHPA membership
- 150 PPG flights after you first got licensed
- 10 flights with a tandem-rated PPG instructor

- At least 300 high flights and 150 hours
- For free-flight tandem - Basic for 2 years and Sport for 12 months
- No careless incidents in the last 6 months
- Pass the tandem theory / skills and show you can brief a passenger

Equipment

- a tandem-rated wing
- a tandem reserve
- a passenger harness and proper attachments
- helmets for both
- an airband radio
- a motor that can lift you + passenger + kit

Question 25: A lighter all-up take-off weight will increase: **(1)**

a) Climb rate.

b) Take-off distance.

c) Power off stall speed

d) Power on stall speed

Question 26: **(1)**

A heavier all-up take-off weight will increase:

a) overall airspeed

b) stall speed

c) take-off run internal cell pressure

d) all of the above

Question 27: **(1)**

The _____ is an imaginary straight line drawn through an airfoil (wing) from the leading edge to the trailing edge.

a) angle of attack.

b) camber fine

c) chord line

d) relative wind line

Question 28: **(1)**

Angle of attack is defined as:

a)The angle between the horizon and the chord line

b) The angle between the horizon and the relative wind

c) The angle between the wing chord line and the vertical tail

d) The angle between the wing chord line and the relative wind

Question 29: **(1)**

A stall occurs when:

a) The critical angle of attack is exceeded. ✓

b) The nose is too high above the horizon.

c) The airspeed gets too low.

d) The engine quits at a bad time.

Question 30: **(1)**

Trimspeed is defined as:

a) flying off the brakes, off speedbar, and trimtabs set to neutral ✓

b) flying off the brakes, on maximum speedbar

c) flying at 50% throttle setting

d) flying as slowly as possible

Question 31: **(1)**

In heavy turbulence it is safest to minimise deflations by flying:

a) Downwind.

b) At as high a speed as possible.

c) At a very slow speed ✓

d) At Trimspeed

Question 32: **(1)**

You are on a head-on collision course with another aircraft in open areas. You should avoid the other aircraft by: ✓

a) Turning to the right

b) turning to the left

c) climbing

d) diving

Question 33: **(1)**

You are flying north in straight and level flight with a True Airspeed of 38km/h. The wind at your altitude is from 180 degrees at 10 km/h. What is your groundspeed?

a) 38 mph

b) 48 mph ✓

c) 28 mph

d) 10 mph

Question 34: **(1)**

A magnetic compass is accurate:

a) With the powered paraglider in any attitude

b) Only when the airplane operated below manoeuvring speed



c) At all times

d) Only when the airplane is operated in straight and level flight at constant speed

Question 35: **(1)**

You would most likely find turbulent flying in conditions near:

a) Stratus clouds

b) Fog

c) Cirrus clouds ✓

d) Cumulus clouds

Question 36: **(1)**

You would likely find turbulence

a) near mountains

b) in the mid afternoon

c) deep inland ✓

d) downwind of buildings

e) all of the above

Question 37: **(1)**

Why does SAHPA exist?

a) to keep the sport self regulating

b) to provide 3rd party aviation insurance

c) to provide a safety orientated licensing system

d) to provide international recognition to the sport ✓

e) all of the above

Question 38: **(1)**

What is the very last safety action to perform prior to starting an engine?

a) check the fuel/oil ratio on the fuel.

b) check for sufficient fuel quantity.

c) do a pre-flight inspection.

d) yell “Clear prop” and verify that the prop area is clear of personnel. ✓

Question 39: **(1)**

Considering density altitude, your motor and wing would perform better on

a) warm humid days.

b) cold humid days.

c) warm dry days.

d) cold dry days.

Question 40: **(1)**

When may you enter a Restricted Area (FAR)?

a) Never!

b) After receiving permission from the appropriate authority.

c) Whenever you wish with extreme caution

d) Only on weekends when the areas are closed.

Question 41: **(1)**

When may you enter a Prohibited Area (FAP)?

a) Never!

b) Only with a written waiver from the CAA Administrator or his designee.

c) With verbal permission from the controlling agency

d) Anytime after official sunset and before official sunrise

Question 42: **(1)**

What frequency is used for emergency communications

a) 126.7 Mhz

b) 124.8 Mhz

c) 130.35 Mhz

d) 121.5 Mhz

Question 43: **(1)**

Take off run distance is increased by:

a) Lighter headwind

b) Higher Altitude

c) Higher Humidity
d) Lower QNH
e) All of the above

Question 44: **(1)**

Carb mixture adjustment should be

a) richened for lower altitudes

B) richened for higher QNH

c) richened for lower humidity

b) All of the above

Question 45: **(1)**
Which of the following is true:

a) below 1500 foot AGL, the altimeter subscale is set to local QNH

b) below 1500 foot AGL, radio your altitude in Feet.

c) above transition altitude, the altimenter subscale is set to QNE

d) When the altimeter subscale has been changed to QNE, radio your altitude in "Flight Level"

e) All of the above

Question 46: **(1)**
Which one is true:
Rate of climb is

A) affected by your throttle setting, unaffected by the windspeed and direction

B) increased in a headwind, reduced in a tailwind

Question 47: **(1)**

Angle of climb is

a) increased in a headwind

b) increased with added power

c) reduced at faster airspeed trim

d) reduced in turns

e) all of the above

Question 48: **(1)**

When Taking off in thermic conditions, you are 10 meters up when you fly into a thermal. The glider Pitches back, your reaction is.

a) Stay on full power and increase break pressure

b) Come to Idle power immediately and pump the breaks

c) maintain 3/4 power and catch the pendulum as the glider surges forward

d) do nothing

e) Scream ✓

Total Marks (150)